

Mahdi Saleh

Munich, Germany

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WORK EXPERIENCE

Senior Applied Scientist Oct 2024 - Present

Zillow Inc, Germany

Working on 3D computer vision and generative AI solutions to enable new user experiences.
Conducting university collaboration with TUM and research internship projects.

Senior Research Engineer July 2024 - Sept 2024

Virtual Staging AI (Acquired By Zillow Group)

Working on developing an instant furniture staging for real estate photography.

PhD Research Scientist Intern Dec 2022 - Apr 2023

Meta Reality Labs, Redmond, WA

Research Associate Sep 2019 - June 2024

Technical University of Munich, Munich

- National AI-Safeguarding Project: 3D object detection and 6DoF pose estimation from 3D data and 3D human pose estimation from RGB camera.
- Researcher on point cloud and mesh processing, 3d pose estimation, graph neural networks, hierarchical understanding, and self-supervised learning.

PhD Research Scientist Intern Jul 2021 - Aug 2021

Facebook Reality Labs, Sausalito, CA

Research Engineer Nov 2017 - Aug 2019

Framos GmbH, Munich

- National IRIS Project: Autonomous robotic container unloading using 3D perception and machine learning
- Leading several projects on reinforcement learning, human pose estimation, reconstruction, and depth estimation.

Augmented Reality Developer Jul 2017 - Nov 2017

AR-Experts, Munich

- Design and implementation of augmented reality training apps for Microsoft HoloLens

Artificial Intelligence Engineer May 2017 - Jul 2017

IBM Watson IoT Center, Munich

- Design and implementation of a wearable helping low vision people in product recognition in supermarkets using deep object recognition and gesture control

Research Developer Aug 2015 - Mar 2017

Framos GmbH, Munich

- Extracting anthropometric data of humans using contour alignment and marker detection
- Pallet dimensioning and 3D reconstruction for logistic pipelines

EDUCATION

Doctor of Philosophy (Ph.D.) - Computer Science

Jan 2019 - Jan 2025

Technical University of Munich, Munich - passed with highest distinction, **summa cum laude**

Thesis title: Hierarchical Representations for 3D Shape Understanding

M.Sc. of Biomedical Computing - Computer Science

Oct 2014 - Oct 2017

Technical University of Munich, Munich

B.Sc. of Electrical Engineering

Sep 2008 - Sep 2013

Iran University of Science and Technology

SKILLS

3D computer vision, sparse view reconstruction, Geometry processing, Deep learning, 3D Detection, and 6D Pose Estimation, Graph Neural Networks, PyTorch, Computer Graphics, Self-supervised Learning, Generative Media, 3D Gaussian Splatting and novel view synthesis

HONORS & AWARDS

CVPR Outstanding Reviewer	2021
Zeidler-Mint Makerspace Scholarship Winner	2016
TUM HackRoboy 1st place winner	2016
Elite Student of among the whole class of electrical engineering	2008

SELECT PUBLICATIONS

CVPR 2026, ConceptPose: Training-Free Zero-Shot Object Pose Estimation using Concept Vectors, L. Kuang, Y. Velikova, M. Saleh, J. Zaeche, DP Paudel, B. Busam

CVPR 2025, ESCAPE: Equivariant Shape Completion via Anchor Point Encoding, B. Bekci, N. Navab, F. Tombari, M. Saleh

CVPR 2022, *Bending Graphs: Hierarchical Shape Matching using Gated Optimal Transport*, M. Saleh, SC. Wu, L. Cosmo, N. Navab, B. Busam, F. Tombari

CVPR 2022, *ZebraPose: Coarse to Fine Surface Encoding for 6DoF Object Pose Estimation*, M. Saleh, Y. Su, T. Fetzer, J. Rambach, B. Busam, N. Navab, D. Stricker, F. Tombari

Neurips 2021, *CoFiNet: Reliable Coarse-to-fine Correspondences for Robust PointCloud Registration*, H. Yu, F. Li, M. Saleh, B. Busam, S. Ilic

3DV Oral 2020, *Graphite: GRAPH-Induced feaTure Extraction for Point Cloud Registration*, M. Saleh, S. Dehghani, B. Busam, N. Navab, F. Tombari

LEADERSHIP EXPERIENCE

BMC Master Program Coordinator TUM	2020 - 2022
Head of IEEE student branch IUST	2011 - 2012
Vice President of Scientific Committee of EE faculty at IUST	2010 - 2011